

A New Gene Expression Dataset for GRNsight: a Web Application for Visualizing Gene Network Models

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<http://dondi.github.io/GRNsight/>



GRNsight Automatically Lays Out Unweighted and Weighted Gene Regulatory Network Graphs

1. File Formats

- Can import and export Excel, SIF, or GraphML files.
- Demo files are provided.

2. Grid Layout

- Grid Layout button allows the users to toggle the graph between a grid layout and a force graph layout.

3. Force Graph Parameter Sliders

- Link distance determines the minimum distance between nodes.
- Nodes have a charge, which repel or attract other nodes.
- Reset functionality sets all parameters to default.
- Locking the parameters prevents any further changes.

4. Node Coloring

- This menu allows users to modify parameters of the node coloring visualization.
- Dataset options are automatically generated from expression data sheets detected in an Excel input workbook.
- A back-end PostgreSQL gene expression database can also be used to color nodes.
- With the addition of data from the current poster there is a total of four datasets available.

5. GRNsight includes options to show or hide the weight values

- Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover of the edges.

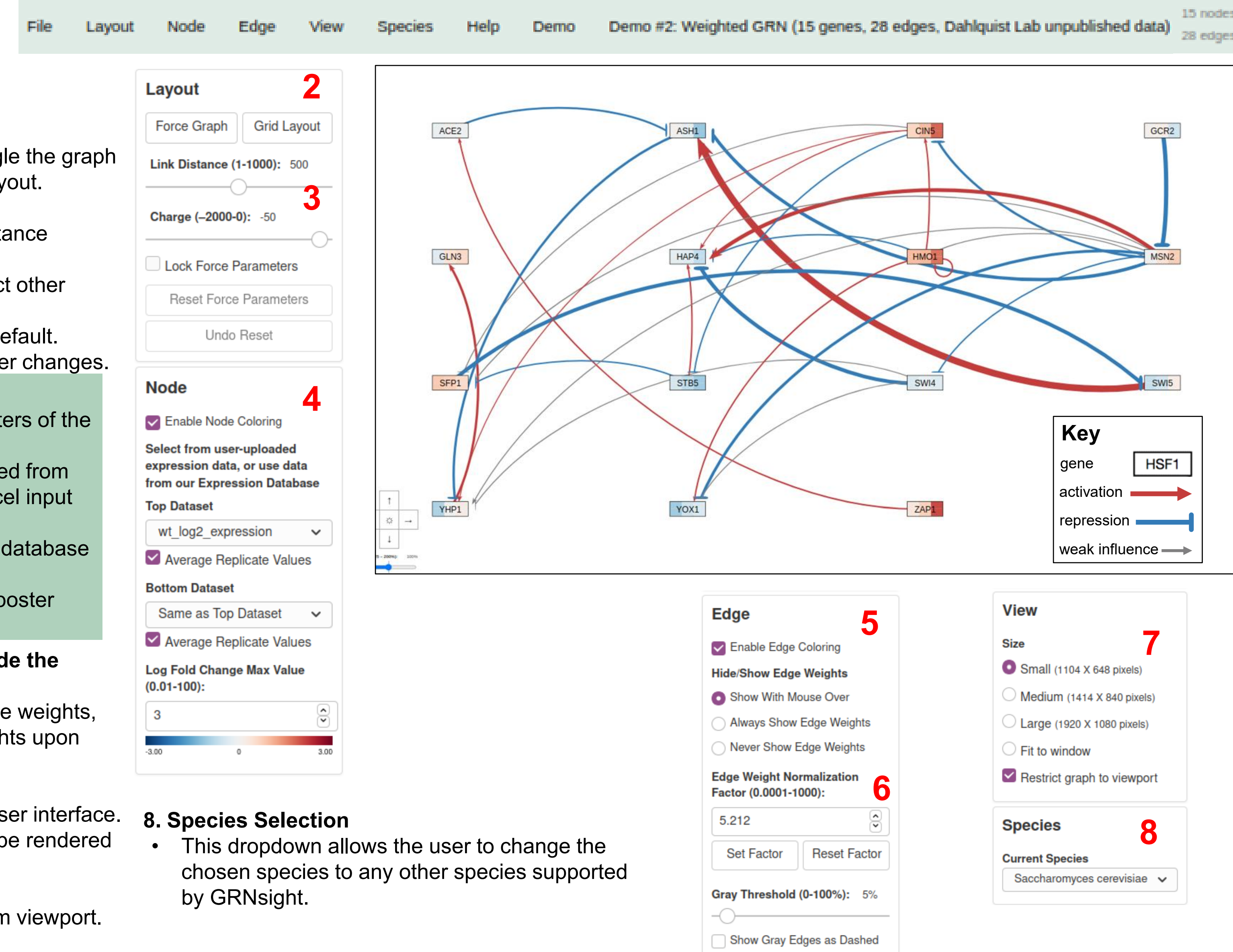
6. Edge Weight Normalization

- Allows user to set normalization factor in user interface.
- Edge thicknesses for different graphs can be rendered on the same scale.

7. Viewport

- Graph bounding box can be separated from viewport.
- Multiple viewport sizes available.
- Zooming and scrolling enabled.

1



8. Species Selection

- This dropdown allows the user to change the chosen species to any other species supported by GRNsight.

Apweiler et al. (2012) Investigated the Effect of Glucose on Gene Expression in Budding Yeast

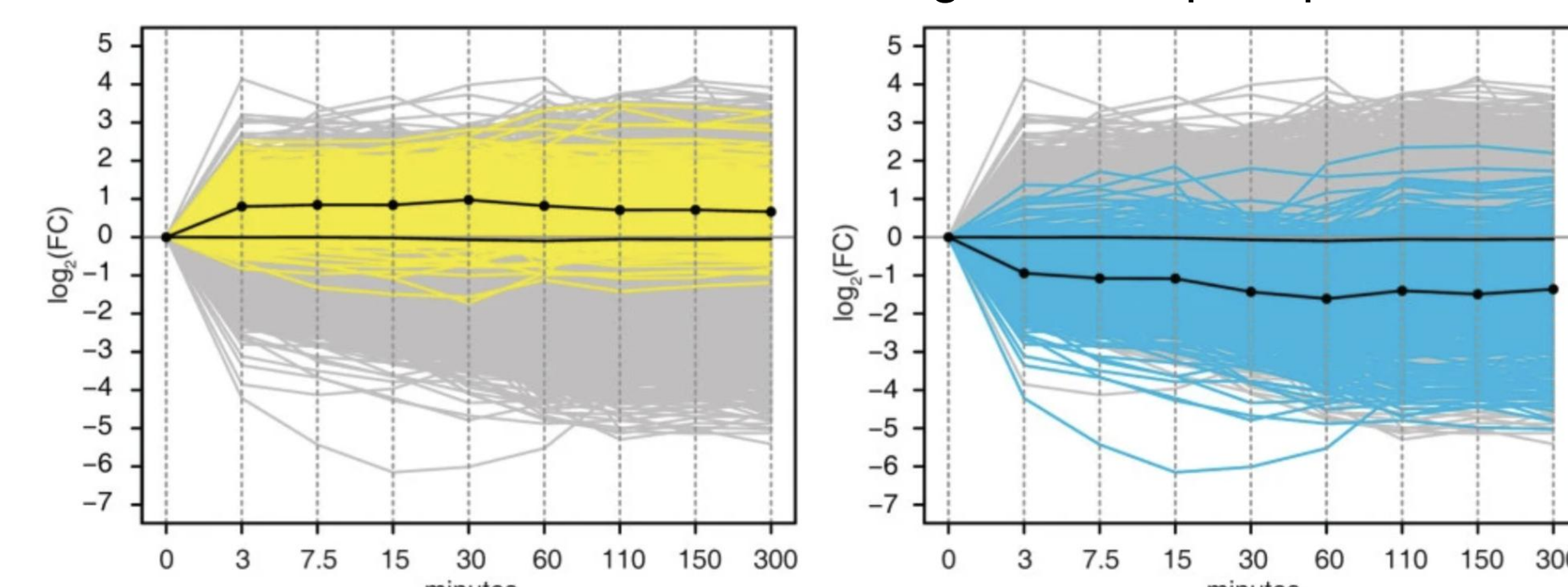
- Apweiler et al. (2012) conducted a time-course experiment measuring the gene expression of *Saccharomyces cerevisiae* Wild Type (WT) cells that were glucose-depleted.
- The figure shows that the transcripts behave in one of two ways, upregulation (yellow) or downregulation (blue).
- The grey portion of the graph demonstrates the nonsignificant transcript expression, and the black line shows the average transcript expression.

Two WT culture were used to inoculate 50mL cultures at an OD₆₀₀ of 0.15 overnight.

Grown for 24 hours to be depleted of glucose.

Samples for expression profiles were taken at 0, 3, 7.5, 15, 30, 60, 110, 150, and 300 minutes after inoculation.

Used to inoculate 500 mL cultures in fresh medium to an OD₆₀₀ of 0.15.



This figure shows the line graph of transcriptional output of glucose-depleted Wild Type cells inoculated into fresh media (Synthetic Complete medium, supplemented with 2% glucose) over a period of 5 hours (Apweiler et al., 2012).

Preparation of Apweiler et al. (2012) Dataset for Addition to GRNsight Database

- For each timepoint, take the average of the duplicate log₂ fold change measures for each gene.
- Convert all the gene IDs into Systematic IDs using the YeastMine database.
- Remove repeated expression data because of alias gene IDs with a custom Python script.



```
sys = re.compile("Y[A-P]([A-Z]{0-9}){0-9}([A-Z]{0-9}){0-9}([A-Z]{0-9}){0-9}")
sgd = re.compile("SGD:[A-Z]{0-9}([A-Z]{0-9}){0-9}([A-Z]{0-9}){0-9}([A-Z]{0-9}){0-9}")

with open("Users/ahmad@n/Desktop/exp_data_cleaning/sys_SGD_ID.txt", "a") as textfile:
    for x, y in dict.items():
        if len(y) > 1:
            sys_list = list(filter(sys.match, y))
            sgd_list = list(filter(sgd.match, y))
            if len(sys_list) == 1: # this should return systematic names
                newlist = sys_list + list(x)
            elif len(sgd_list) == 1: # this should return systematic names
                newlist = sgd_list + list(x)
            else:
                anotherlist = newlist + list(x)
                sys_list = sys_list + list(x)
                sgd_list = sgd_list + list(x)
            textfile.write(sys_list)
            textfile.write(sgd_list)
            textfile.write("\n")
        elif len(sgd_list) == 1: # this should return SGD names
            newlist = sgd_list + list(x)
            textfile.write(newlist)
            textfile.write("\n")
        elif len(sys_list) == 1: # this should return sys names
            newlist = sys_list + list(x)
            textfile.write(newlist)
            textfile.write("\n")
```



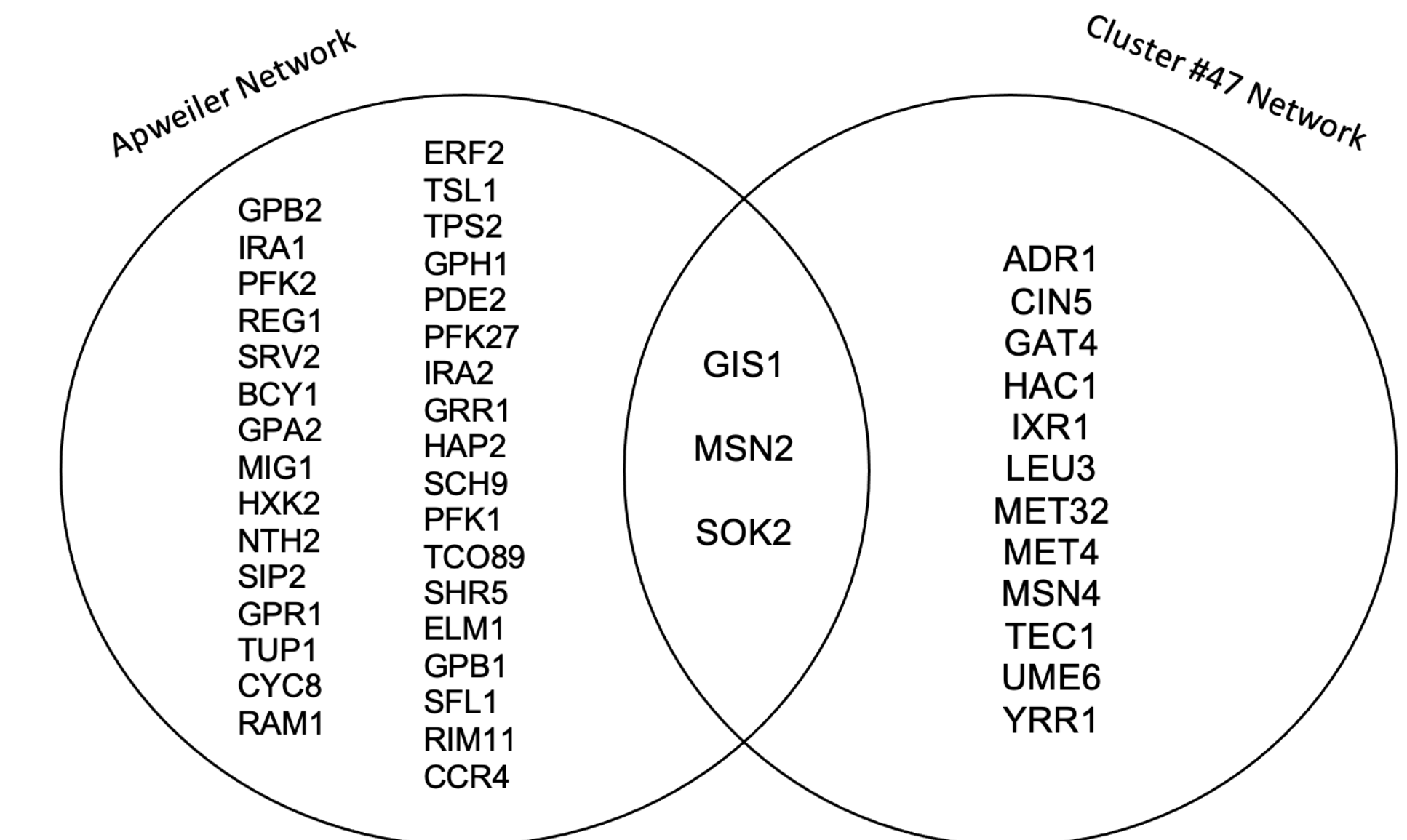
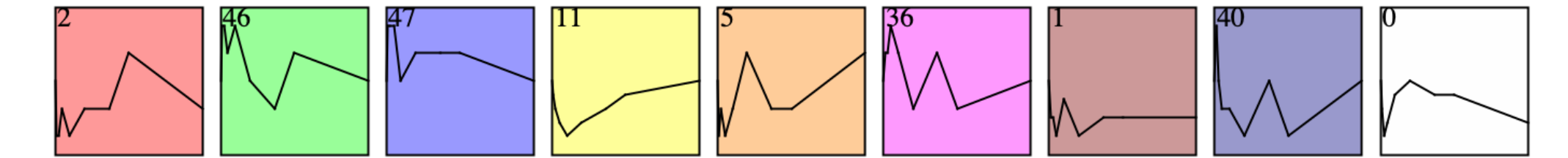
Systematic Name	Aliases	Systematic Name	Standard Name	SGD ID
YAR042W	YAR044W, OSH1	YAR042W	SWH1	SGD:S000000081

STEM Clustering to Observe Gene Expression Patterns

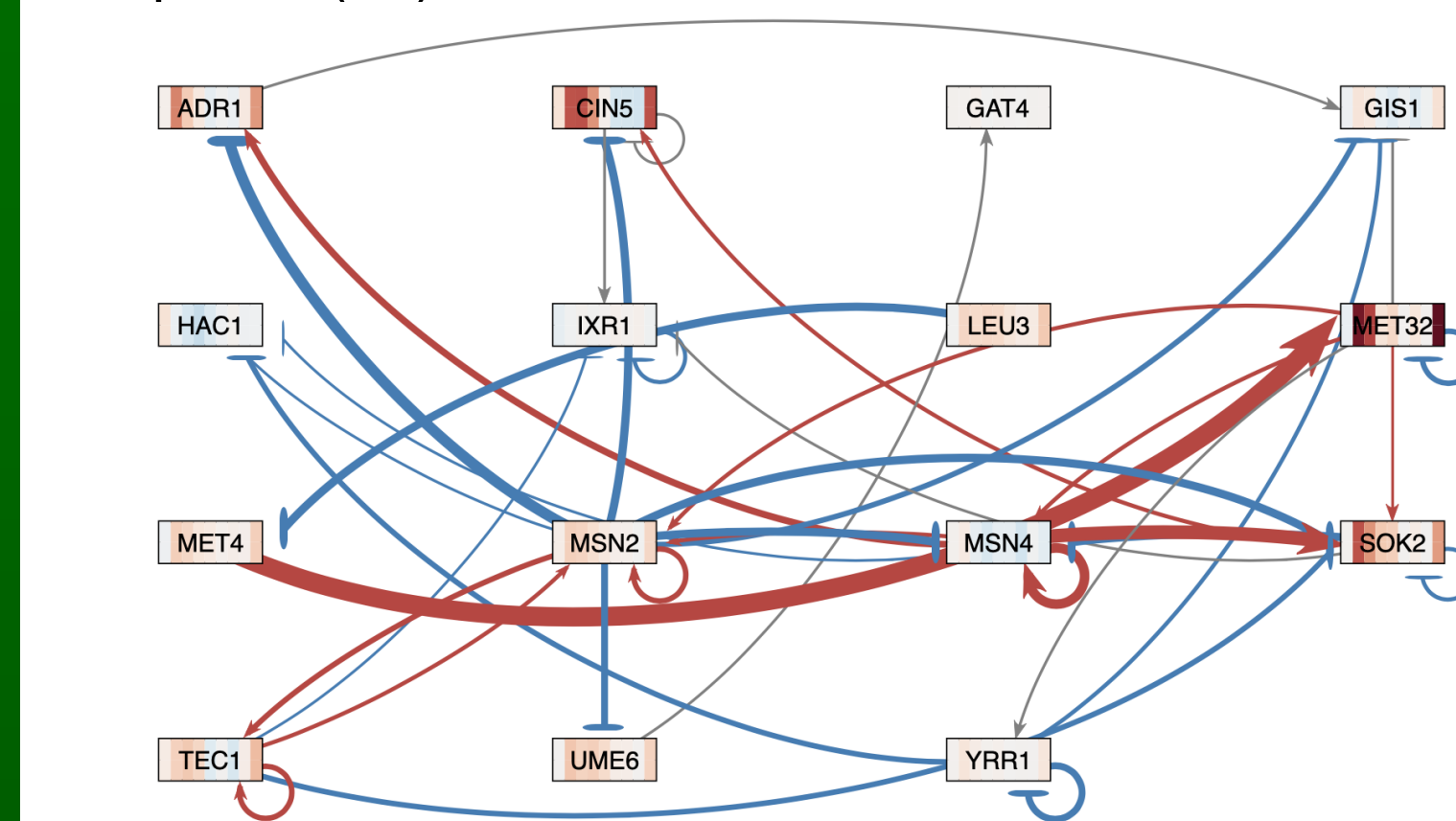
Clustering analysis using STEM

- Run the STEM clustering method to observe gene expression patterns.
- It found 8 significant profiles with up- and down-regulation patterns.
- The first four significant profiles were chosen to be analyzed further.
- The profiles from STEM are consistent with the conclusions from the Apweiler et al. (2012) paper.
- Find the candidate transcription factors responsible for the expression patterns observed in the four clusters.
- Generate a gene regulatory network for the first four significant clusters and the top 15 transcription factors using YEASTRACT.
- Visualize the networks with GRNsight to check that the network is completely connected.
- Generate another gene regulatory network with the same genes using the *Saccharomyces* Genome Database.
- The Venn diagram shows there are 3 common transcription factors between our profile #47 network and the network reconstruction from the Apweiler et al. (2012) paper.
- There were no common transcription factors for the most significant profile (#2).

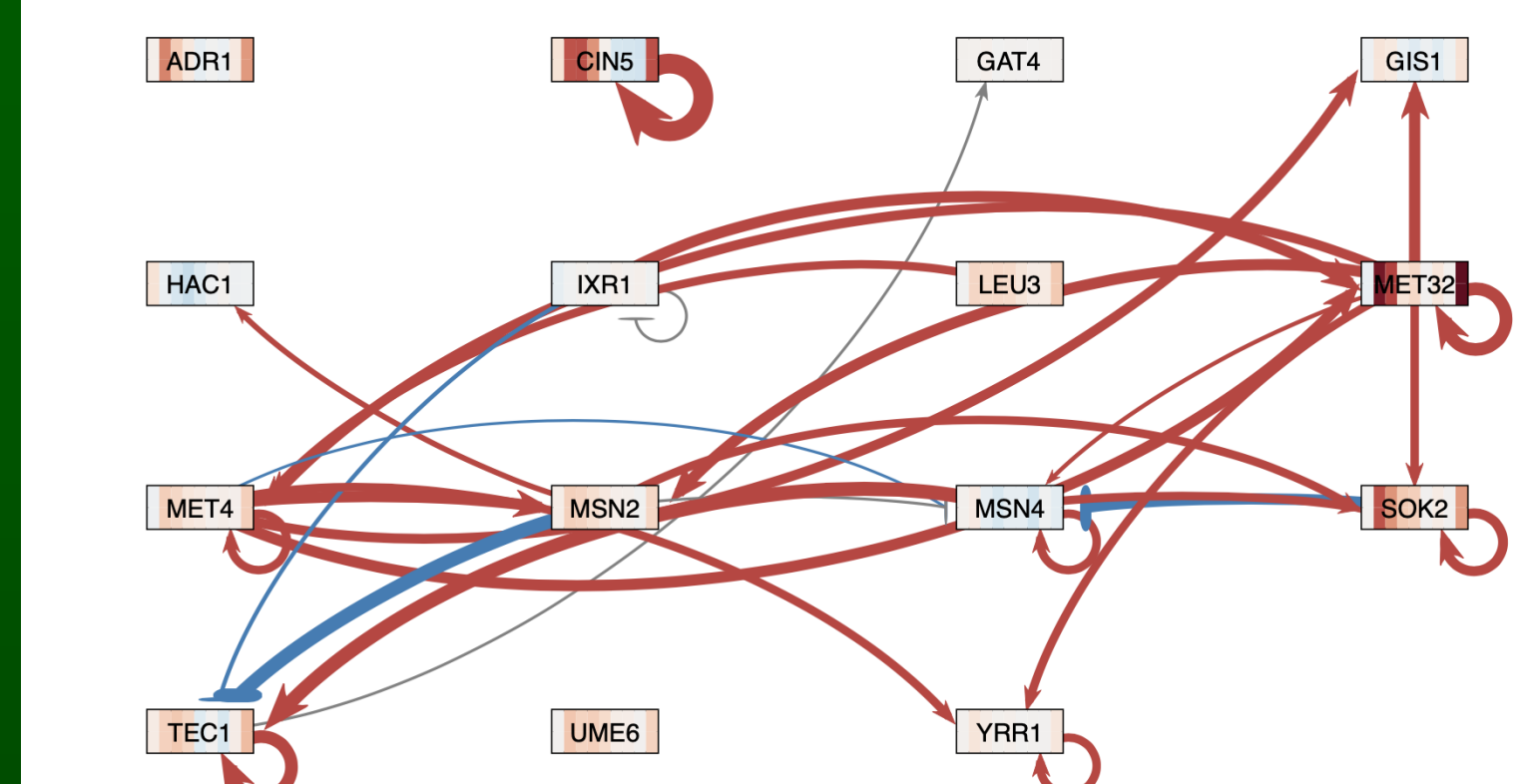
Clusters ordered based on number of genes and profiles ordered by significance (default)



- Apweiler et al. (2012) performed a hierarchical network reconstruction method for network analysis with a larger number of genes, working with both of their clusters, while our networks are smaller and based on a single cluster.

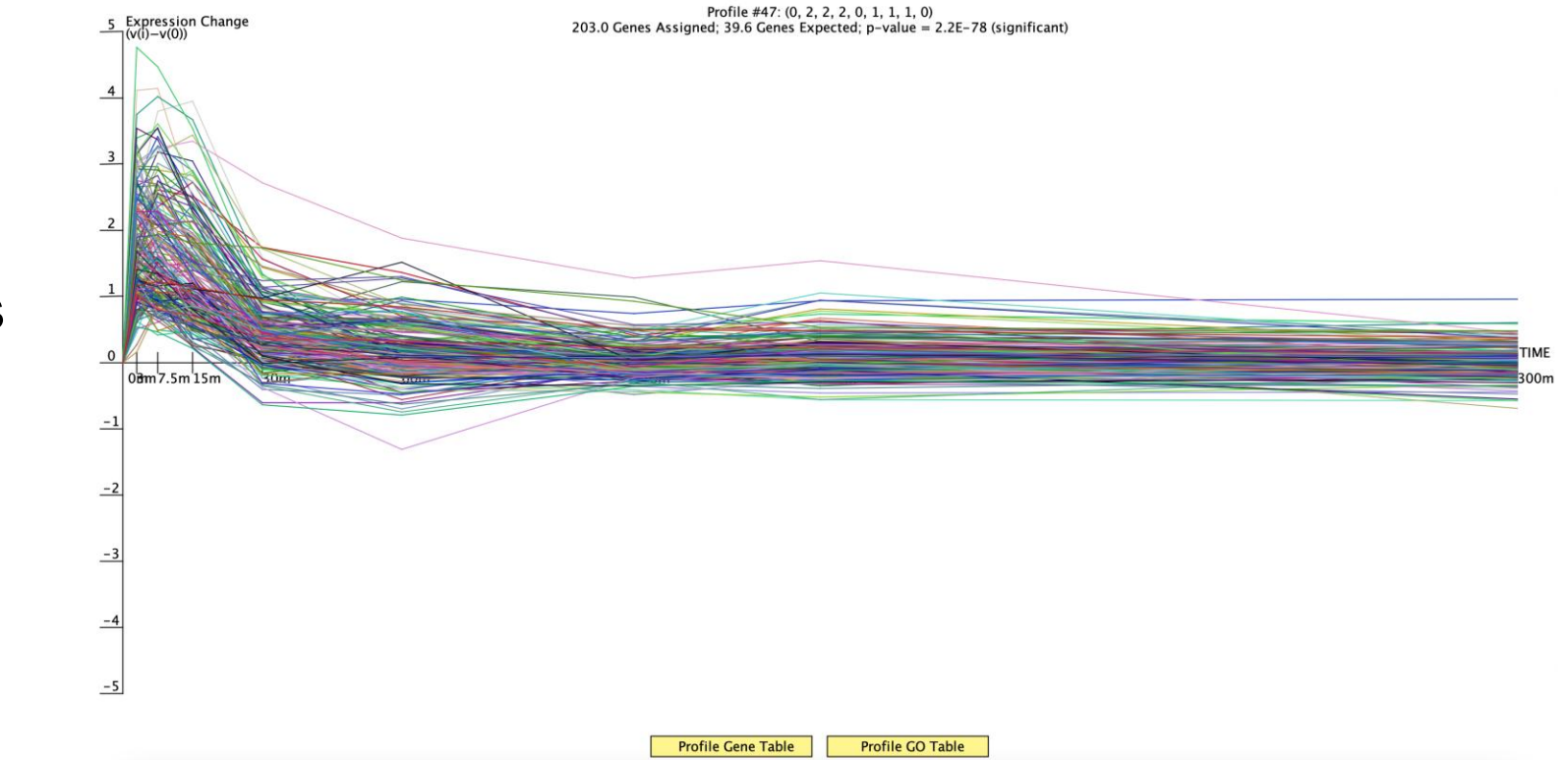


Gene Regulatory Network of profile 47 from YEASTRACT Database with 38 edges. The weight of the edges were calculated using GRNmap.

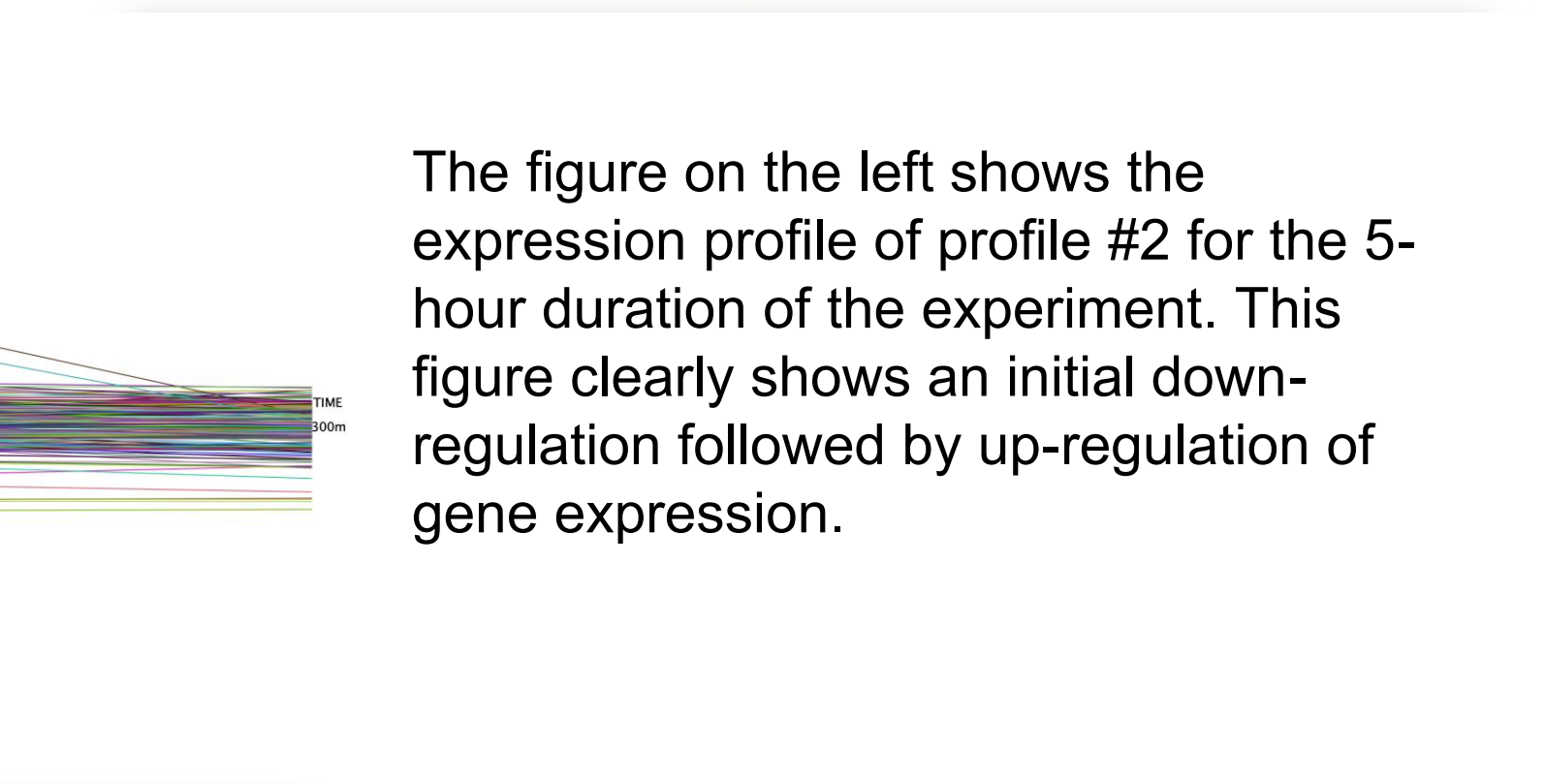


Gene Regulatory Network of profile 47 from *Saccharomyces* Genome Database with 30 edges. The weight of the edges were calculated using GRNmap.

The figure on the right shows the expression profile of profile #47 for the 5-hour duration of the experiment. This figure clearly shows an initial up-regulation followed by down-regulation of gene expression.



The figure on the left shows the expression profile of profile #2 for the 5-hour duration of the experiment. This figure clearly shows an initial down-regulation followed by up-regulation of gene expression.



Availability

- GRNsight is free and open to all users and there is no login requirement.
- Web site content is available under the Creative Commons Attribution Non-Commercial Share Alike license.
- GRNsight code is available under the open-source BSD license.
- Usage is being tracked through Google Analytics.
- Current version on Master: V5.1.0

New Users: 7,346

File Uploads: 7,811

Sessions: 18,091

Pageviews: 38,472

Future Directions

- Load Apweiler et al. (2012) dataset into PostgreSQL.
- Run GRNmap with alternative parameters to obtain more accurate weight estimation of the direction strength and weight of the activation/repression of genes.
- Work on outstanding bugs and feature enhancements to the graph display and user interface.

References

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- Yeasttract: <http://www.yeasttract.com/index.php>
- SGD: <https://www.yeastgenome.org>
- GRNmap: <http://dondigithub.github.io/GRNmap/>
- GRNsight: <https://dondi.github.io/GRNsight/>
- GRNsight's GitHub: <https://github.com/dondi/GRNsight>

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